Installing LAMP (Linux + Apache + MySQL + PHP/Perl/Python) is a fairly widely used option for configuring servers with Ubuntu. There are a large number of applications that are open source and written using the LAMP application stack. The popular ones are: Wiki encyclopaedias, content management systems (CMS), and management applications such as phpMyAdmin.

An important advantage of LAMP is its flexibility in the selection of databases, Web servers and scripting languages. The current replacement for MySQL is PostgreSQL and SQLite. Python, Perl and Ruby can be replaced by PHP. And nginx, Cherokee and Lighttpd are alternatives to Apache.

For quick LAMP installation, use Tasksel, which is a Debian/Ubuntu tool that installs multiple dependent packages on your system as a single task.

LAMP can be used for:

- If necessary, an environment for Web development and testing applications that are written for it. This is both for your own applications and for any CMS you need. You can run it on Windows and on Ubuntu.
- Creating a high-performance Web server on a virtual VPS server or on a dedicated server for hosting your projects.
- A server for version control systems.
- Self-learning administration.
- Economic purposes, to create your own server.

Why does LAMP security matter?

The security of your business starts with your database — the larger the business, the more the data that needs to be stored and processed. Servers are used to store a lot of data. For this reason, they are the main target of attackers.

Today, there are many ways to save your data and prevent hackers from taking over your confidential information, including special software, firewalls and antivirus programs. It is very important to properly configure them and professionally monitor events in order to protect data and prevent its theft.

A subset of information security or data security is ensuring the security of your data. With more reliance on computers, there are a number of potential threats to the data you store. Data may be lost due to a system crash, corrupted by a computer virus, deleted, or altered by a hacker. A simple user error can cause a file to be overwritten or deleted. In addition, lost devices, such as a tablet or smartphone, can cause your data to fall into the wrong hands.

Ways to secure Linux

Automatic updates: The LAMP stack is based on Linux and the whole open source community is working on its improvements. On Ubuntu VPS, all security updates and patches are available for automatic installation as soon as they reach the Ubuntu repositories; so make sure you set the system to automatically install these if you are concerned about security. In case this feature is not enabled on the server and you do not install the latest updates and patches manually, you are putting your server at risk of hacking.

To enable unattended automatic updates, you must install the *unattended-upgrades* package:

```
sudo apt-get install unattended-
upgrades
```

To configure which package categories will be automatically updated, you must edit the */etc/apt/apt.conf.d/50unattended-upgrades* file.

Configuring the firewall: On Linux Ubuntu, the UFW firewall configuration tool is disabled by default. You can enable it with one command: *sudo ufw enable*. After activation, open access to OpenSSH and Apache and if necessary, allow access to other services with the *sudo ufw allow [port number]* command. Allow access to basic services like OpenSSH and Apache:

```
sudo ufw allow 22
sudo ufw allow 80
sudo ufw allow 443
```

Enabling access to other services is pretty easy. Just replace the port number in the above examples with the port number of the service you want to provide access to, and that's it. The firewall rules will remain active even after the system is restarted.

Disabling unused services: If you have active services that you don't use, you can simply disable them. For example, if you have a service like Dovecot running on a server and you are not using it, stop and disable the service using the following commands:

```
sudo systemctl stop dovecot.service
sudo systemctl disable dovecot.service
```

Ways to secure Apache

Fail2ban: This is a utility for a Web server on Ubuntu, designed to register unauthorised intrusions. The software monitors log files and registers failed login attempts and their number, and detects automatic attacks. When the utility recognises an attempt to compromise the system, the IP from which the requests were made is blocked. In iptables, a new chain is simply added and the attack on the server can be considered a failure. To install Fail2ban, run the following command:

```
sudo apt-get install fail2 ban
```

Create a copy of the default configuration file, so that you can safely make changes:

```
sudo cp /etc/fail2ban/jail.conf /etc/
fail2ban/jail.local
```

Edit the *jail.local* file:

```
sudo nano /etc/fail2ban/jail.local
The [SSHD] block should look like this:
[sshd]
enabled  = true
```